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Efficient Frontier Using Quantum Computing

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Introduction

The Efficient Frontier represents the optimal balance between risk and return in portfolio management.

Traditional methods face challenges when handling discrete constraints (sector limits, cardinality, fixed weights), making the problem computationally expensive.

By reformulating portfolio optimization into a QUBO (Quadratic Unconstrained Binary Optimization) problem, we can leverage Quantum Computing techniques such as QAOA (Quantum Approximate Optimization Algorithm) and Quantum Annealing.

Quantum systems explore multiple portfolio combinations simultaneously through superposition and interference, enabling faster, scalable, and more realistic optimization than classical approaches.

This approach helps construct a Quantum-Optimized Efficient Frontier, providing clearer risk–return trade-offs and supporting better investment decisions in real-world financial markets.

Proposed Methodology

Step-1: Problem Formulation (Efficient Frontier)

Classical Setup

The goal of portfolio optimization is to find the best allocation of funds across a set of assets to achieve an optimal balance between risk (variance of portfolio returns) and expected return.

Objective: $$\min_{\omega} \lambda \omega^T \sum \omega - (1-\lambda)\mu^T \omega$$

Constrains:

Budget: $1^T \omega = 1$ (all money invested)

No Short Selling: $\omega \geq 0$

Realistic Constraints:

Real portfolios often require additional real-life constraints, making the problem combinatorial and thus computationally challenging:

Cardinality constraint: Choose exactly K assets out of N.

Discrete weights: Investments can only be multiples of fixed steps (e.g., 5% increments).

Sector restrictions: Limits on investments by sector (e.g., maximum 3 stocks in financials).

Step-2: QUBO Transformation

Quantum solvers need a Quadratic Unconstrained Binary Optimization (QUBO) form: $\min_{x \in \{0,1\}^n} x^T Q x$

Where: $x_i = 1$ asset i is selected;

Q encodes the combined efforts of risk, return, and constraints.

The QUBO Objective becomes:

$$f(x) = \lambda x^T \sum x - (1-\lambda)\mu^T x + M(1^T x - K)^2$$

Risk term: $\lambda x^T \sum x$ penalizes risky portfolios.

Return term: $-(1-\lambda)\mu^T x$, rewards high return.

Penalty: $M(1^T x - K)^2$,ensures exactly K assets are chosen.

M is a large constant controlling penalty strength.

By encoding portfolio optimization this way, the problem suits quantum techniques for combinatorial optimization.

Step 3: Mapping QUBO to Quantum Computers

A. Quantum Approximate Optimization Algorithm (QAOA)

- Maps the QUBO problem into a quantum Hamiltonian using Pauli-Z operators.
- The quantum circuit alternates between applying:

- Cost Unitary : $U_c(\gamma)$ pushes states towards low-energy (optimal) portfolios.
- Mixer Unitary : $U_M(\beta)$ explores various portfolio combinations.
- After p layers, measures the quantum state to sample candidate portfolios.
- Classical optimizer fine-tunes angles γ, β to minimize expected cost.
- Sweeps over λ to obtain full efficient frontier with quantum solutions.

B. Quantum Annealing (QA)

- Expresses the problem as an Ising model energy function:

$$E(s) = \sum_i h_i s_i + \sum_{i<j} J_{ij} s_i s_j$$

- Starts in an easy-to-prepare ground state, then slowly morphs the system to the target Hamiltonian reflecting portfolio optimization.
- System settles into low-energy states corresponding to optimal asset selections.
- Particularly effective on specialized QA hardware (e.g., D-Wave).

Step 4: Building the Efficient Frontier

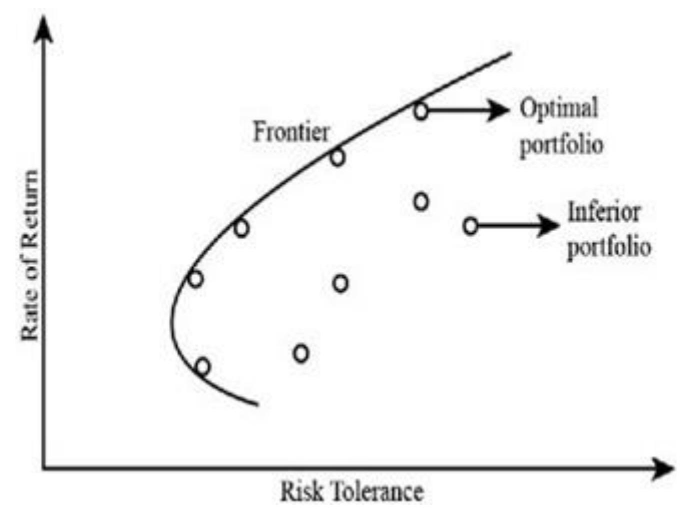
1.Define a grid of λ values between 0 and 1 controlling risk-return preference.

2. For each λ:

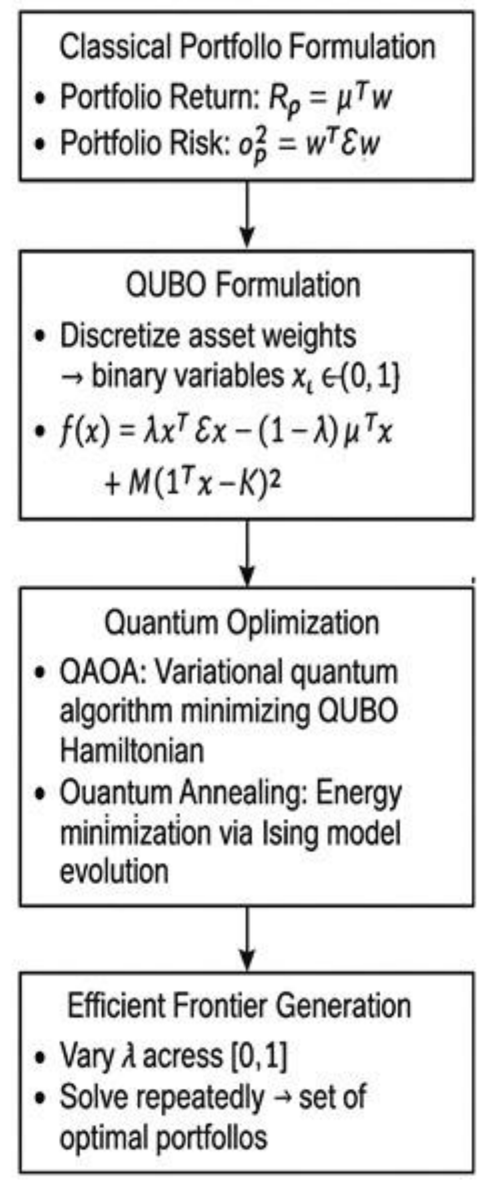
- Formulate the QUBO representation.
- Run on quantum solver (QAOA or Annealer).
- Obtain best portfolio x^*
- Calculate Return $R = \mu^T x^*$
- Calculate Risk .$\sigma^2 = x^{*T} \sum x^*$.

3. Collect (σ,R) points for all λ.

4. Plot the Quantum Efficient Frontier, showcasing optimal trade-offs.



Block Diagram



Required Resources

- By considering top 50 (Nifty Companies) for 3 years data and taking O H L C (open, high, low and close stock).
- There will be 252 working days per year for market, considering all these parameters,
- The feature size is: 252 * 3 * 4 * 50= 1,51,200 features .
- Required Qubits= log2(1,51,200) = 17.2 Qubits for features

Advantages

Explores Complex Portfolios Efficiently → Quantum superposition enables simultaneous evaluation of many asset combinations.

Handles Discrete Constraints → Accurately incorporates real-world factors like cardinality, sector restrictions, and fixed investment steps.

Faster Frontier Construction → Quantum solvers sweep through multiple λ values more quickly than classical optimization.

Clearer Risk–Return Trade-offs → Produces a quantum-optimized efficient frontier for better investment decision-making.

Scalable to Larger Portfolios → Quantum algorithms reduce computational bottlenecks faced by classical solvers.

Hybrid Future-Ready Approach → Combines classical & quantum strengths, aligning finance with next-gen quantum hardware.

Conclusion

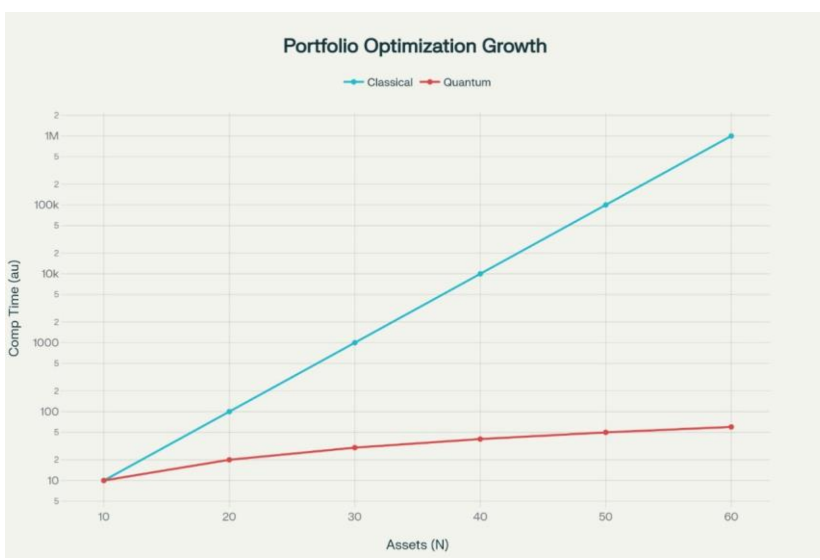
The Efficient Frontier using Quantum Computing demonstrates how classical portfolio optimization challenges can be reformulated into QUBO models solvable on quantum hardware.

Techniques such as QAOA and Quantum Annealing enable faster exploration of risk–return trade-offs, even under real-world constraints like cardinality, sector limits, and discrete weights.

By leveraging superposition and interference, quantum systems build a quantum-optimized efficient frontier, offering clearer investment insights, scalability to large portfolios, and future-ready solutions.

This approach positions finance at the forefront of the quantum revolution, paving the way for hybrid quantum–classical strategies in real-world portfolio management.

Graphs



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